

Detecting What LLMs Don't Know

Investigating black-box uncertainty estimation in RAG systems: An exploratory engineering case study

OSU MIND & MACHINE ALIGNMENT SUMMIT 2026

Exploring the Trust Gap

How do we identify hallucination signatures in production-ready generative AI systems when model internals remain strictly hidden behind black-box APIs?

| The Black-Box Dilemma

The API Constraint

Production pipelines often interact with commercial models as absolute black boxes. These systems expose **strictly raw text outputs**, precluding access to logits, layer activations, or token probabilities.

Case Study Objective

We explore whether an **output-only uncertainty estimation technique** can consistently identify behavioral patterns correlated with incorrect and ungrounded generation.

The STASI Dataset

-  **Purpose:** A publicly available dataset, used as-is (not benchmarked or built in-house) to isolate and profile distinct hallucination behaviors.
-  **Archetypes:** Segmented into lookup-based facts, semantic ambiguity, and unresolvable missing-context entries.
-  **Engineering Utility:** Serves as an ideal sandboxed environment to map how uncertainty signals shift across distinct retrieval situations.

Direct Lookup

Direct retrieval of clear factual records

Ambiguity

Multiple valid interpretations of target query

Missing Context

Completely unresolvable search spaces

The EigenEval Workflow



1. Stochastic Sampling

Generate 10 candidate responses from the target model under high temperature settings to trigger semantic variation.



2. Pairwise NLI

Evaluate semantic entailment and contradiction between all generated candidate pairs to calculate mutual logical agreement.



3. Graph Analysis

Build a semantic agreement graph where nodes are responses and edge weights reflect pairwise consistency scores.

System Configuration

System Component	Technology Specification	Role in Pipeline
Target LLM	Gemma-2-2B-it (Open-Weights Instruction Model)	Generates candidate response corpus stochastically
Exploratory Dataset	STASI Dataset	Simulates direct, ambiguous, and missing-context prompts
Hybrid Retrieval	BM25 (Sparse) + BGE-M3 (Dense Vector Embeddings)	Fetches contextual background records for target prompts
Context Reranking	Cross-Encoder Reranker Architecture	Filters and prioritizes top retrieved context matches
Uncertainty Evaluator	DeBERTa-v3 NLI Model	Executes fast pairwise entailment and contradiction checking
Stochastic Generation	10 Passes (High temperature setting)	Provides sample distribution to form semantic networks

Quantitative Stress-Test Results

Pilot stress test, n=40 — 10 direct, 20 missing-context, 10 ambiguous. Directional signal, not a statistically powered evaluation.

DIRECT LOOKUP · n=10

10/10 (100%)

correct answers

2/10 (20%)

flagged uncertain anyway (false positives)

MISSING CONTEXT · n=20

12/20 (60%)

of answers fabricated

7/12 (58%)

of fabrications caught (rest converged confidently)

AMBIGUITY · n=10

8/10 (80%)

silently picked one answer

0/10 (0%)

spontaneously offered multiple valid answers

Across all three categories: the uncertainty score tracks self-consistency, not correctness.

| Key Limitation: Consistent $\neq$ Correct

“EigenEval detects self-disagreement — not when a question truly has multiple right answers.”

5/20 (25%)

missing-context questions produced confident, self-consistent fabrication — the method never saw a signal

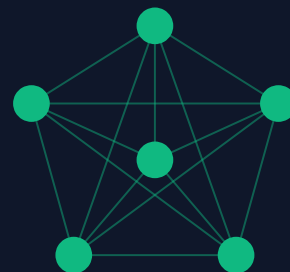
8/10 (80%)

ambiguous questions were silently resolved to one answer, with no disagreement across samples

Pilot stress test (n=40) — both failures share one cause: the uncertainty signal tracks decoding stability, not ground truth.

Semantic Graph States

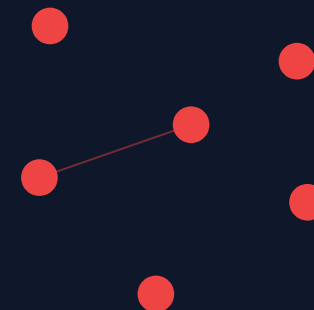
- 🔗 **Supported State:** Tight cluster, high consensus density. Eigenvalue matches approach maximum bounds.
- 🔗 **Ambiguous State:** Splitting network profiles reflecting distinct factual branches or name variations.
- 🚫 **Confabulated State:** Highly fragmented, random graph topology indicating complete context mismatch.



Supported State



Ambiguous State



Confabulated
State

Deployment Considerations



Computational Latency

Sampling 10 responses per query and computing pairwise NLI transitions multiplies inference cost — overhead compounds directly with sample count and context length.



Real-Time Serving Conflict

Running the full pipeline inside the live request path adds latency directly to response time — there is no free place in the architecture to hide the extra generations.



Threshold Calibration

The eigenvalue/consensus score is continuous, not a decision — there is no principled cutoff for "too uncertain," and an untuned threshold risks high false-positive or false-negative rates.

| Engineering Future Directions

Failure-Mined Gate Model

Stress-test the production model to mine confident-fabrication and divergence cases, then distill them into a lightweight open-source gate that triggers abstention or human review — no per-query sampling cost at serve time.

Failure-Informed Prompts

Restructure prompts from mined failure patterns — explicit multi-record disclosure for ambiguity, explicit abstention for missing context. Cheapest fix to try first; strong on ambiguity, only partial on fabrication.

Thank You